



Lecture 2: Conditioning and Bayes' rule

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1. Review of probability models
2. Conditional probability
3. Important tools
4. Recitation
5. Summary



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- Sample space Ω
 - Mutually exclusive & Collectively exhaustive
 - Right granularity



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- Allocation of probabilities to events
 - $\mathbf{P}(A) \geq 0$
 - $\mathbf{P}(\Omega) = 1$
 - If $A \cap B = \emptyset$, then $\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B)$.
 - * If $A_1, A_2, \dots$ are **disjoint** events, then
$$\mathbf{P}(A_1 \cup A_2 \cup \dots) = \mathbf{P}(A_1) + \mathbf{P}(A_2) + \dots$$

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- Problem solving “recipe”
 - Specify sample space
 - Define probability law
 - Identify event of interest
 - Calculate...

- A narrow view: a branch of math
 - Axioms $\implies$ theorems

Theorem: “Frequency” of event A “is” $\mathbf{P}(A)$

- Are probabilities frequencies?
 - $\mathbf{P}(\text{coin toss yields heads}) = 1/2$
 - $\mathbf{P}(\text{Trump wins 2024 US Election}) = 0.1$
 - $\mathbf{P}(\text{a piece of equipment aboard the space shuttle fails}) = 10^{-8}$
- Probability models are a framework for describing uncertainty
 - Use for consistent reasoning
 - Use for predictions, decisions
- **Model vs Reality: tradeoff between accuracy and tractability**

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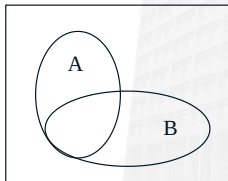
- Partial information is available
- $\mathbf{P}(A \mid B)$ = **conditional probability** of A given that B occurred
 - B is our new universe: suppose that the outcome is in event B

Definition

Assuming $\mathbf{P}(B) \neq 0$, the conditional probability

$$\mathbf{P}(A \mid B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)}$$

$\mathbf{P}(A \mid B)$ undefined if $\mathbf{P}(B) = 0$



- For a fixed event B , $\mathbf{P}(A \mid B)$ can be verified as a legitimate probability law on the new universe
 - **Axiom 1:** holds straightforwardly
 - **Axiom 2:**

$$\mathbf{P}(\Omega \mid B) = 1$$

- **Axiom 3:** If $A_1, A_2, \dots$ are **disjoint** events, then

$$\mathbf{P}(A_1 \cup A_2 \cup \dots \mid B) = \mathbf{P}(A_1 \mid B) + \mathbf{P}(A_2 \mid B) + \dots$$

- Discrete uniform law: finitely many and equally likely

$$\mathbf{P}(A \mid B) = \frac{\# \text{ of elements of } A \cap B}{\# \text{ of elements of } B}$$

Example 1: Coin toss

- Toss a fair coin three times
- Let A be the event that more Heads than Tails
- Let B be the event that 1st toss is Head
 - $\mathbf{P}(A \mid B) =$



Example 2: Die roll

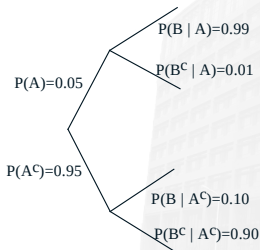
- Let B be the event: $\min(X, Y) = 2$
- Let $M = \max(X, Y)$
 - $\mathbf{P}(M = 1 \mid B) =$
 - $\mathbf{P}(M = 2 \mid B) =$

Y = Second roll

4				
3				
2				
1				
	1	2	3	4

X = First roll

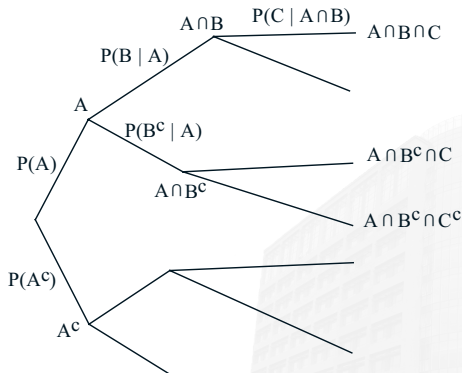
- Modeling experiments with sequential character: from conditional to unconditional probability
- Event A : Airplane is flying above
- Event B : Something registers on radar screen
 - $\mathbf{P}(A \cap B) =$
 - $\mathbf{P}(B) =$
 - $\mathbf{P}(A \mid B) =$



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- Tree-based sequential description of an experiment
- $\mathbf{P}(A \cap B \cap C) = \mathbf{P}(A) \cdot \mathbf{P}(B \mid A) \cdot \mathbf{P}(C \mid A \cap B)$



- Extension to more events

Example

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$$A_i = \{\text{the } i\text{th card is not a heart}\} \quad i = 1, 2, 3.$$

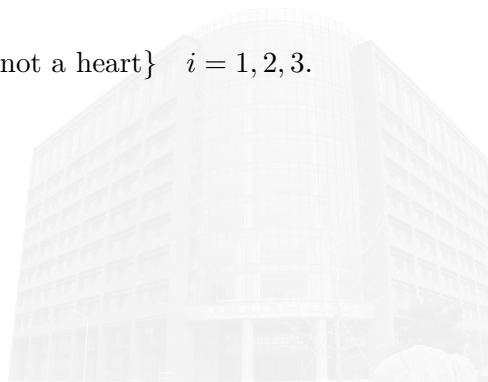
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- $\mathbf{P}(A_2 \mid A_1) = \frac{38}{51}$

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- $P(A_2 | A_1) = \frac{38}{51}$
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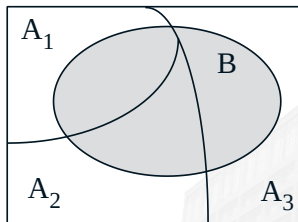
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- $P(A_3 | A_1 \cap A_2) = \frac{37}{50}$
- $P(A_1 \cap A_2 \cap A_3) = \frac{39}{52} \cdot \frac{38}{51} \cdot \frac{37}{50}$

- Complicated event: Divide and conquer
- Partition of sample space into A_1, A_2, A_3
- Have $\mathbf{P}(B \mid A_i)$, for every i



- One way of computing $\mathbf{P}(B)$

$$\begin{aligned}\mathbf{P}(B) &= \mathbf{P}(A_1 \cap B) + \mathbf{P}(A_2 \cap B) + \mathbf{P}(A_3 \cap B) \\ &= \mathbf{P}(A_1)\mathbf{P}(B \mid A_1) + \mathbf{P}(A_2)\mathbf{P}(B \mid A_2) + \mathbf{P}(A_3)\mathbf{P}(B \mid A_3)\end{aligned}$$

Theorem

Let $A_1, \dots, A_n$ be disjoint events that form a partition of the sample space and assume that $\mathbf{P}(A_i) > 0$ for all i . Then for any event B , we have

$$\mathbf{P}(B) = \sum_{i=1}^n \mathbf{P}(A_i \cap B) = \sum_{i=1}^n \mathbf{P}(A_i) \mathbf{P}(B \mid A_i)$$

Remarks:

- The probability of B is a weighted average of its conditional probability under each scenario
- Each scenario is weighted according to its prior probability
- Useful when $\mathbf{P}(B \mid A_i)$ is known or easy to derive

Example

- You roll a fair four-sided die. If the result is 1 or 2, you roll once more but otherwise, you stop.
- What is the probability that the sum total of rolls is at least 4?

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- What is the probability that the sum total of rolls is at least 4?

Solution:

- Define the events

$$A_i = \{\text{the result of first roll is } i\} \quad i = 1, 2, 3, 4.$$

and

$$B = \{\text{the sum total is at least 4}\}$$

- Therefore,

$$\mathbf{P}(B \mid A_1) = \frac{1}{2} \quad \mathbf{P}(B \mid A_2) = \frac{3}{4} \quad \mathbf{P}(B \mid A_3) = 0 \quad \mathbf{P}(B \mid A_4) = 1$$

- Note that $\mathbf{P}(A_i) = \frac{1}{4}$

$$\mathbf{P}(B) = \frac{1}{4} \cdot \left(\frac{1}{2} + \frac{3}{4} + 0 + 1 \right) = \frac{9}{16}$$

- “Prior” probabilities $\mathbf{P}(A_i)$
 - initial “beliefs”
- We know $\mathbf{P}(B \mid A_i)$ for each i
- Wish to compute “posterior” probability $\mathbf{P}(A_i \mid B)$
 - revise “beliefs”, given that B occurred

$$\begin{aligned}\mathbf{P}(A_i \mid B) &= \frac{\mathbf{P}(A_i \cap B)}{\mathbf{P}(B)} \\ &= \frac{\mathbf{P}(A_i)\mathbf{P}(B \mid A_i)}{\mathbf{P}(B)} \\ &= \frac{\mathbf{P}(A_i)\mathbf{P}(B \mid A_i)}{\sum_j \mathbf{P}(A_j)\mathbf{P}(B \mid A_j)}\end{aligned}$$

Bayes' Rule

Let $A_1, \dots, A_n$ be disjoint events that form a partition of the sample space and assume that $\mathbf{P}(A_i) > 0$ for all i . Then for any event B such that $\mathbf{P}(B) > 0$, we have

$$\begin{aligned}\mathbf{P}(A_i | B) &= \frac{\mathbf{P}(A_i)\mathbf{P}(B | A_i)}{\mathbf{P}(B)} \\ &= \frac{\mathbf{P}(A_i)\mathbf{P}(B | A_i)}{\sum_{j=1}^n \mathbf{P}(A_j)\mathbf{P}(B | A_j)}\end{aligned}$$

Remarks:

- Relates conditional probabilities of the form $\mathbf{P}(A_i | B)$ with conditional probabilities of the form $\mathbf{P}(B | A_i)$
- Often used in inference: effect $B \leftrightarrow$ cause A_i

- If a person has caught COVID-19, the results of nucleic acid tests are positive with probability 0.95. If the person does not have the disease, the test results are negative with probability 0.95. Suppose that a random person drawn from a certain population has probability 0.001 of having the disease.
- Given that a person has just tested positive in his nucleic acid test, what is the probability of having COVID-19?

- If a person has caught COVID-19, the results of nucleic acid tests are positive with probability 0.95. If the person does not have the disease, the test results are negative with probability 0.95. Suppose that a random person drawn from a certain population has probability 0.001 of having the disease.
- Given that a person has just tested positive in his nucleic acid test, what is the probability of having COVID-19?

Solution:

- Define the events

$$A = \{\text{the person has the disease}\}$$

and

$$B = \{\text{the test results are positive}\}$$

- $\mathbf{P}(A \mid B)$, is

$$\begin{aligned}\mathbf{P}(A \mid B) &= \frac{\mathbf{P}(A)\mathbf{P}(B \mid A)}{\mathbf{P}(A)\mathbf{P}(B \mid A) + \mathbf{P}(A^c)\mathbf{P}(B \mid A^c)} \\ &= \frac{0.001 \cdot 0.95}{0.001 \cdot 0.95 + 0.999 \cdot 0.05} \\ &= 0.0187\end{aligned}$$

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1. We roll two fair 6-sided dice. Each one of the 36 possible outcomes is assumed to be equally likely.
 - (a) Find the probability that doubles are rolled.
 - (b) Given that the roll results in a sum of 4 or less, find the conditional probability that doubles are rolled.
 - (c) Find the probability that at least one die roll is a 6.
 - (d) Given that the two dice land on different numbers, find the conditional probability that at least one die roll is a 6.

2. A coin is tossed twice. Is it right that the event of two heads is at least as likely if we know that the first toss is a head than if we know that at least one of the tosses is a head. Does it make a difference if the coin is fair or unfair?

3. You enter a chess tournament where your probability of winning a game is 0.3 against half of the players (call them type 1), 0.4 against a quarter of the players (call them type 2), and 0.5 against the remaining quarter of the players (call them type 3). You play a game against a randomly chosen opponent.

- (a) What is the probability of winning?
- (b) Suppose that you win. What is the probability that you had an opponent of type 1?

4. **The Monty Hall Problem.** You are told that a prize is equally likely to be found behind any one of three closed doors in front of you. You point to one of the doors. A friend opens for you one of the remaining two doors, after making sure that the prize is not behind it. At this point, you can stick to your initial choice, or switch to the other unopened door. You win the prize if it lies behind your final choice of a door. Consider the following strategies:

- (a) Stick to your initial choice.
- (b) Switch to the other unopened door.
- (c) You first point to door 1, and your friend always chooses to open door 2 if the prize is behind door 1. If door 2 is opened, you do not switch. If door 3 is opened, you switch.

Which is the best strategy?

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- Conditional probability

$$\mathbf{P}(A \mid B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)}$$

- Important tools
 - Multiplication rule
 - Total probability theorem
 - Bayes' rule
- Readings: Section 1.3-1.4
- Assignment: Problem Set 2
- Next lecture: Independence

